

Artefact: Discussion Forum Contributions (Intelligent Agents)

This artefact collects my contributions to the three collaborative discussions of the Intelligent Agents module, with links to the original forum threads for review, and summarises what I argued and what I learned in each.

Collaborative Discussion 1: Agent-Based Systems

Forum: <https://www.my-course.co.uk/mod/forum/view.php?id=1344226>

Initial post: <https://www.my-course.co.uk/mod/forum/discuss.php?d=364899>

Summary post: <https://www.my-course.co.uk/mod/forum/discuss.php?d=366391>

My initial post examined why agent-based systems have risen with organisational complexity and the demand for autonomous decision-making, analysing applications in healthcare (intelligent patient monitoring), cybersecurity (adaptive threat detection), and the life sciences (agent-based modelling of biological and epidemiological systems), and comparing reactive, deliberative BDI, and multi-agent approaches. Peer replies deepened the thread considerably: the point I valued most was the accountability gap in multi-agent systems, where outcomes emerge from collective interactions rather than a single identifiable source. My summary post consolidated the discussion's conclusion, which became a theme of my whole module: the success of agent-based systems depends not only on autonomy and technical capability but on transparency, governance, explainability, and appropriate human involvement.

Collaborative Discussion 2: Agent Communication Languages

Forum: <https://www.my-course.co.uk/mod/forum/view.php?id=1344253>

Initial post: <https://www.my-course.co.uk/mod/forum/discuss.php?d=368484>

Summary post: <https://www.my-course.co.uk/mod/forum/discuss.php?d=372736>

This discussion compared agent communication languages such as KQML with direct method invocation. I explored how performatives, ontologies, and facilitator agents allow genuinely independent agents to coordinate, and argued that flexibility comes at a price: every performative an agent accepts is also something another party can exploit. In my peer responses I proposed concrete mitigations for both the performance and the security concerns raised by colleagues, including hybrid architectures that reserve method invocation for critical deterministic functions,

predefined ontologies and schema validation to limit runtime interpretation, and audit logging to detect abnormal behaviour, connecting the classical literature to modern incidents such as indirect prompt injection against LLM-integrated applications. What I learned here fed directly into my project work: communication between agents is a governance question and an attack surface, not just a technical convenience.

Collaborative Discussion 3: Deep Learning

Forum: <https://www.my-course.co.uk/mod/forum/view.php?id=1344282>

Initial post: <https://www.my-course.co.uk/mod/forum/discuss.php?d=374390>

Summary post: <https://www.my-course.co.uk/mod/forum/discuss.php?d=377906>

My initial post assessed the ethical and social issues raised by generative deep learning models through a cybersecurity lens. I argued that the hardest problem is the dual-use tension: the capability that helps attackers craft convincing phishing and malicious code is inseparable from the one that supports defenders in malware analysis and incident response, so it cannot be released to one side only. I concluded that the ethical issue lies less in the technology than in its governance, pointing to risk-based frameworks such as the EU AI Act and the NIST AI Risk Management Framework. The discussion sharpened my view that responsible deployment is a property of the whole socio-technical system, a position I later applied when evaluating the emergent behaviours of my own agent implementation.

What this artefact demonstrates

Across the three discussions I engaged with peers by building on their arguments, researching beyond the module materials, and supporting every position with referenced literature, from foundational agent papers to current security research and regulation. Together they evidence my critical and analytical thinking about agent-based systems, my awareness of the legal, social, ethical, and professional issues surrounding autonomous systems, and my development in communicating complex technical arguments clearly in an academic setting.